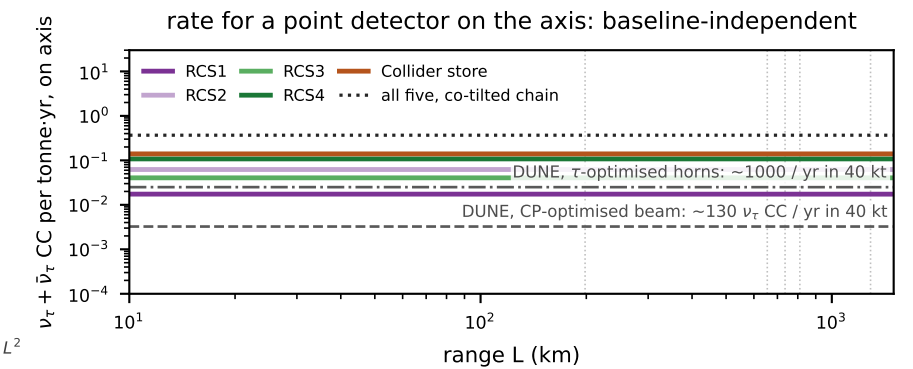
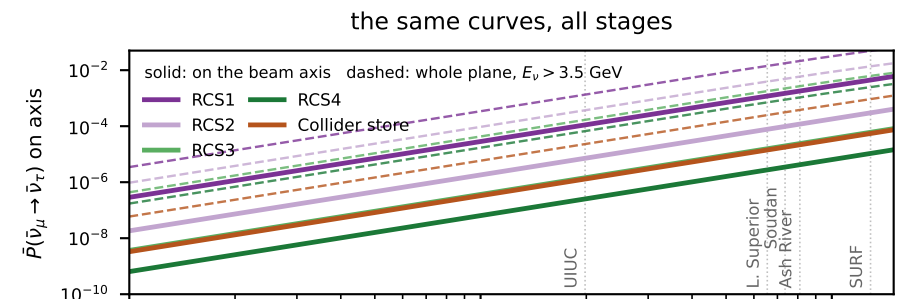
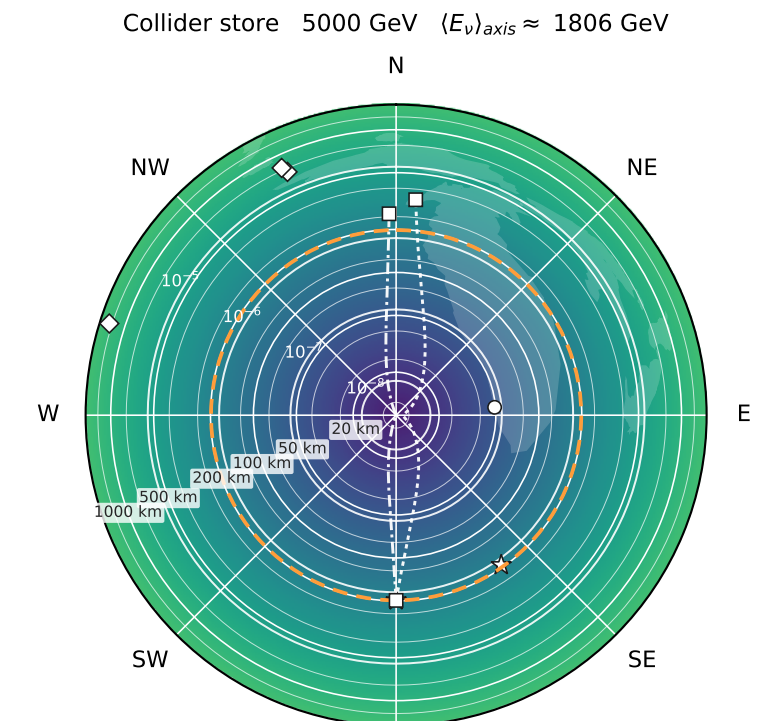
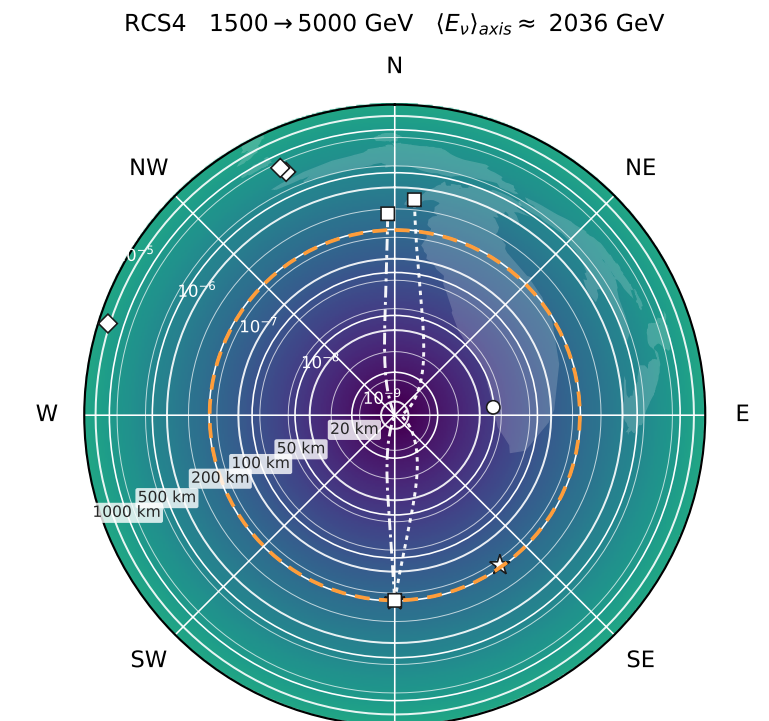
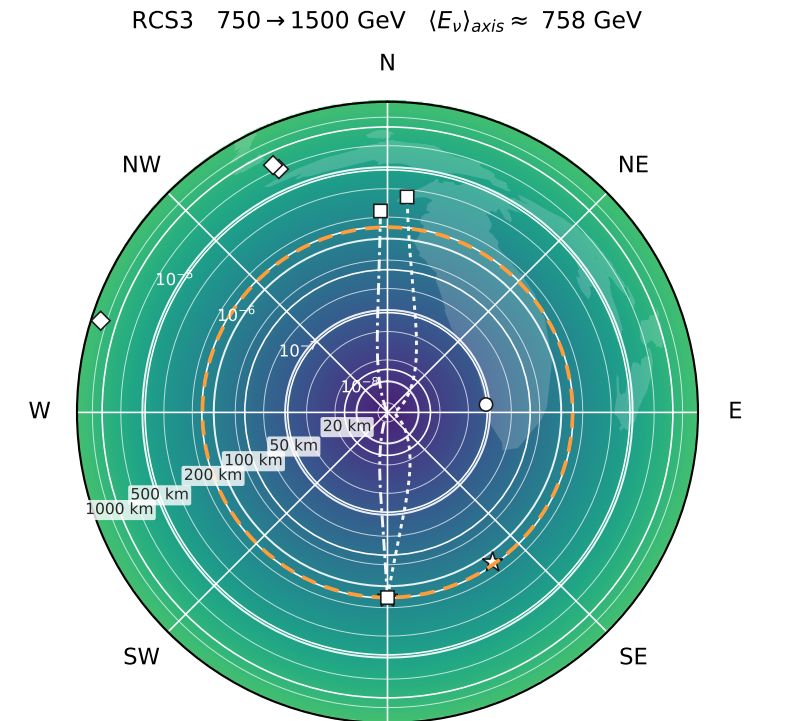
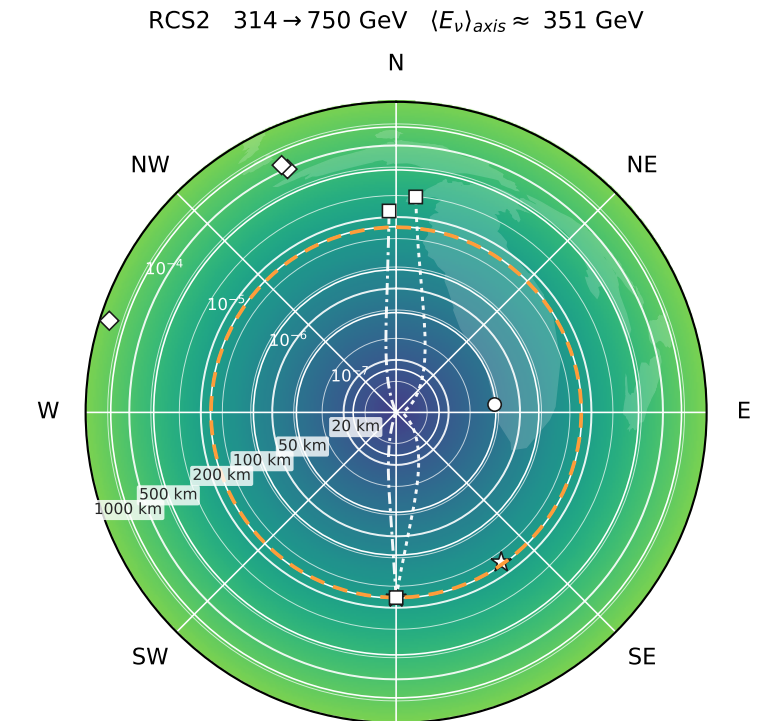
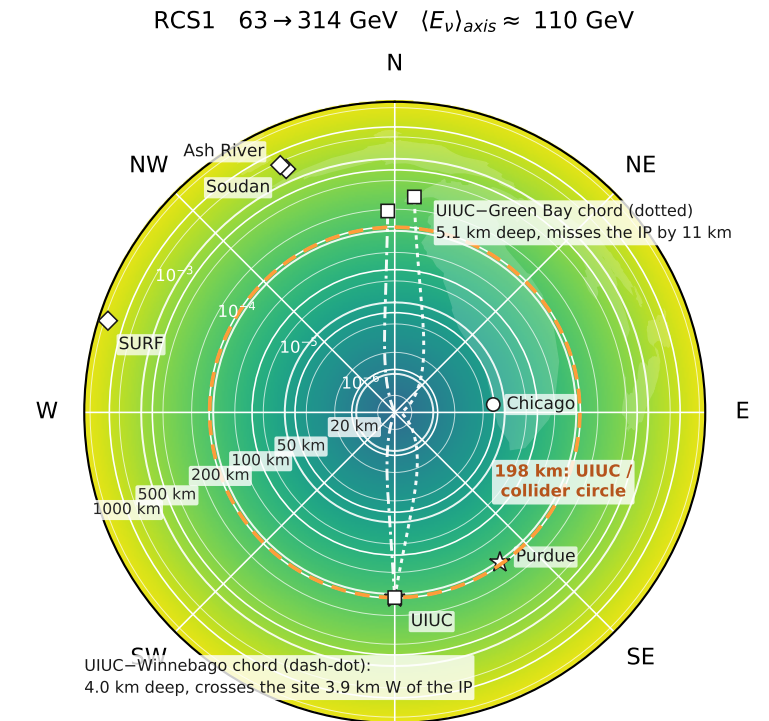


Tau appearance around the compass, stage by stage

colour = fraction of the stage's $\bar{\nu}_\mu$ flux that has become $\bar{\nu}_\tau$ by the range where the beam surfaces at that bearing (it depends on range only: rings). White: lakes. Orange dashed: the 198 km circle the collider and the co-tilted chain exit on.



Point spectra: the RCS pencils are $1/\gamma$ -sharp, so a point on the axis sees the hard forward spectrum ($\langle E_\nu \rangle = 0.7 E_\mu$); the collider plume is smeared to $\sigma_\theta = 7/\gamma$, so its axis sees the whole-plume spectrum ($\langle E_\nu \rangle = 0.35 E_\mu$, cut at $E_\nu/E_\mu = 1/50$) at 1/101 of the pencil density. Rates: north-aimed decays per cycle from chain_timing.json, 5 Hz, 1.2×10^7 s/yr, both signs, CSMS CC cross-sections, τ -threshold suppression; on-axis flux $\propto 1/L^2$ and $\bar{P} \propto L^2$ cancel, so the rate is flat in L . DUNE lines: TDR vol. II §4.1.1.3 ν_τ CC interactions per year before detector efficiency (130 CP-optimised, ~ 1000 τ -optimised) in the 40 kt fiducial far detector at 1300 km, per tonne. White dotted / dash-dot: the UIUC–Green Bay and UIUC–Lake Winnebago chords (two_ends.py).